Mahmoud Hany Mazroa Game Developer

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Profile

Passionate Game Developer dedicated to blending technical execution with narrative vision to craft immersive experiences. A collaborative problem-solver skilled in thoughtful level design and rapid prototyping, seeking to build high-quality game systems in a professional studio.

Education

Bachelor of Computer and Information Sciences, Specialization in Digital Multimedia, 10/2021 – 07/2025
Ain Shams University and University of East London Cairo, Egypt
GPA: 3.148 | Rank: 2nd

9-Month Post-Graduate Diploma - Game Programming Track, 10/2025 – 08/2026
Information Technology Institute (ITI) Smart Village, Cairo, Egypt

Experience

Game Development Instructor, *Kodland* 08/2026 – Present
London, UK (Remote)
- Teach students aged 10 to 12 in game development, Lua scripting, and 3D level design using Roblox Studio in live remote classes
- Mentor students through complete development cycles in implementing game mechanics and interactive logic from concept to playtesting, and debug their scripts in real time

Unity Developer (Contract), *National Bank of Egypt* 05/2026 – 08/2026
Cairo, Egypt
- Developed a resource management mobile game "Cyber Salvage" targeting financial literacy for young audiences

Languages

Arabic
Native

English
Fluent (IELTS 8.0)

French
Conversational

Skills

Programming Languages
- C
- C++
- C#
- Lua

Core Concepts
- Object-Oriented Programming (OOP)
- Data Structures & Algorithms
- Design Patterns
- Dependency Injection (DI)

Version Control
- Git / GitHub
- Unity Version Control (Plastic SCM)

Game Engines & Graphics
- Unity Engine
- Unreal Engine
- Roblox Studio
- SFML
- OpenGL
- WebGL

Game Design & Theory
- Level Design
- Visual Communication
- Rapid Prototyping

Tools
- XR Interaction Toolkit
- Meta XR SDK
- ARCore
- OpenXR
- VContainer
- UI Toolkit

Personal Skills
- Communication
- Leadership
- Presentation Skills
- Research
- Technical Mentorship

Graduation Projects

Cyber Salvage (Mobile, Unity) | ITI Graduation Project, 05/2026 – 08/2026
Commissioned by the National Bank of Egypt (NBE)

- **Architecture:** Built core gameplay loops and factory meta-progression in C#, using VContainer-based Dependency Injection to decouple systems and optimize memory.
- **Economy:** Balanced a dual-loop economy between arena resource collection and simulated market/factory upgrades. *Unity, C#, uGUI, VContainer, Dependency Injection, Mobile Optimization*

The Learning Lens (VR/AR, Unity) | BSc Graduation Project, 10/2024 – 07/2025
Team Leader & Unity/Backend Developer
ali-hy/the-learning-lens at dev
Shokryy/the-learning-lens-ar

- **Spatial UX & Mechanics:** Designed interactive assembly simulations and spatial UX in Unity, guiding users through hands-on industrial and medical training.
- **Full-Stack Integration:** Built backend REST services in .NET and EF Core, bridging live server data with Unity XR components. *Unity, C#, .NET, EF Core,Meta SDK, AR Core, Python*

Projects

Two Bows Off Frame | KSA National Day Game Jam, *WebGL* 09/2026
KSA Game Jam
2D stealth-platformer chasing an escaped cat through three uniquely styled paintings, each granting a distinct traversal ability, while avoiding detection from gallery visitors' shadows

- **Rank: 3rd place and best innovation award**
- Built all gameplay systems, player abilities, stealth detection, level and UI management.
- Handled level design across all three paintings and wrote and voiced the narrative.

Night Shift Taxi | Brackeys Game Jam 2026.2, *WebGL* 08/2026
A comedic 2D Unity taxi game where you navigate a city with no map, only contradictory directions from strangers, judging who's lying to survive the shift and climb the leaderboard.

- Built the game's core night manager, trip loop, and dawn progress systems.

LoadNExplode | GMTK Game Jam 2026, *WebGL* 07/2026
A chaotic score-chaser where players gather massive crowds of NPCs to maximize their blast radius within a strict time limit.

- Developed core gameplay systems, including NPC spawning, aim-based player abilities, and leaderboard integration.

VR Wave Shooter, *(Unity, XRIT)* 04/2026
Designed core gameplay loops including dynamic round progression, scalable enemy spawning, and a strict ammo economy to balance player difficulty.

Subject Null - VR Puzzle Game, *(Unity, XRIT)* 02/2026
Designed levels, puzzles, and core gameplay mechanics including dynamic mesh slicing and distinct vision states for a narrative driven VR game. Additionally, implemented Unity Version Control to facilitate seamless project iteration.

ESCCTRL 2D Atmospheric Platformer, *SFML* 12/2025
eslam135/ESCCTRL

- Designed intricate levels focusing on shape-shifting mechanics and tight spatial navigation within a 3-day game jam.
- Created a dynamic environment using deep parallax layering to react to player movement.